C. MICHAEL HAYNES

Postdoctoral Scholar, Georgia Institute of Technology
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POSITIONS HELD

Georgia Institute of Technology, School of Earth and Atmospheric Sciences Atlanta, GA
Graduate Research Assistant *August 2022 - Present*

- Developed a modular, scalable, and highly distributed computational model to study energetic ion precipitation onto icy moons' atmospheres and their production of Energetic Neutral Atoms (ENAs)
- Applied this model at Callisto, Ganymede and Europa to investigate the intensity and morphology of the emitted ENA flux in preparation for the JUICE mission (en route)
- Devised and deployed a "backtracing" model to efficiently generate high-fidelity synthetic ENA images; applied this tool and coupled it to a hybrid model at Europa, Callisto, Ganymede, and Titan
- Published three first-authored papers and delivered 14 conference presentations describing this work
- Extensively contributed to projects that adapt hybrid simulations to study Mercury's interaction with ICMEs and the imprint of Pluto's induced magnetosphere on its heliospheric ion environment

Georgia Institute of Technology, School of Earth and Atmospheric Sciences Atlanta, GA
Undergraduate Research Assistant *November 2021-July 2022*

- Developed a tracing tool for charge exchange of energetic magnetospheric ions in Europa's atmosphere
- Results of this work were shown in a first-authored presentation at the 2022 MOP conference

Georgia Institute of Technology, School of Physics Atlanta, GA
Undergraduate Research Assistant, Center for Nonlinear Sciences *May 2021-February 2022*

- Derived a system of PDEs describing multi-species evolution of temperature and concentration during different stages of a Carbon Capture process
- Responsible for quarterly technical progress reports delivered to the Department of Energy

Georgia Institute of Technology, School of Mathematics Atlanta, GA
Undergraduate Research Assistant, REU on Dynamical Systems *May 2020-December 2020*

- Analytically calculated Lyapunov spectra and invariant manifolds of a hyperbolic dynamical system

Georgia Institute of Technology, School of Physics Atlanta, GA
Undergraduate Research Assistant, Center for Nonlinear Sciences *May 2019-December 2019*

- Fabricated and soldered circuit components for a high-precision Hall-effect sensor to automatize measurement of the magnetic field for an experimental setup that emulates quasi-2D turbulence

EDUCATION

Georgia Institute of Technology *August 2022 - November 2026*
Doctor of Philosophy (Ph.D.), Planetary & Space Physics Graduation: December 2026
Minor, Physics GPA: 4.0
Advisor: Sven Simon

Georgia Institute of Technology *August 2018 - May 2022*
B.S., Physics, Concentration: Astrophysics High Honors
Minor, Mathematics GPA: 3.52

PUBLISHED JOURNAL PAPERS

3 First-Authored Publications (8 Total). Citations: 42 (Google Scholar). H-index: 3

8. **Induction Signals from Europa's Subsurface Ocean: Contributions of Time Variations in the Plasma Interaction**
   Lucas Liuzzo, Sven Simon, **C. Michael Haynes**, Andrew R. Poppe, *J.Geophys. Res. (Space Physics)*, 131(11), e2026JA036129, doi: 10.1029/2026JA036129, 2026

7. **Modeling the Detectability of Energetic Heliospheric Ions at Pluto During the New Horizons Flyby**
   Randall Ruch, Sven Simon, Peter Kollmann, **C. Michael Haynes**, *J.Geophys. Res. (Space Physics)*, 131(5), e2026JA035261, doi: 10.1029/2026JA035261, 2026

6. **Compression of Mercury's Dayside Magnetopause to the Surface: a Three-Dimensional Model of Magnetospheric Structure and Dynamics**
   Georg Glebe, **C. Michael Haynes**, Sven Simon, *J.Geophys. Res. (Space Physics)*, 131(3), e2025JA034907, doi: 10.1029/2025JA034907, 2026

5. **Emission of Energetic Neutral Atoms From Ganymede's Magnetosphere-Atmosphere Interaction**
   C. Michael Haynes, Sven Simon, Lucas Liuzzo, *J.Geophys. Res. (Space Physics)*, 130(10), e2025JA034469, doi: 10.1029/e2025JA034469, 2025b
   This article was selected as the cover story of the JGR issue (October 2025: Vol. 130(10)); C. Michael Haynes is credited with the cover figure.

4. **Dynamics of Energetic Heliospheric Ions in Pluto's Induced Magnetosphere**
   Randall Ruch, Sven Simon, **C. Michael Haynes**, *J.Geophys. Res. (Space Physics)*, 130(1), e2024JA033548, doi: 10.1029/2024JA033548, 2025

3. **Constraints on the Observability of Energetic Neutral Atoms from the Magnetosphere-Atmosphere Interactions at Callisto and Europa**
   C. Michael Haynes, Tyler Tippens, Sven Simon, Lucas Liuzzo, *J.Geophys. Res. (Space Physics)*, 130(1), e2024JA033391, doi: 10.1029/2024JA033391, 2025a
   This article was selected as the cover story of the JGR issue (January 2025: Vol. 130(1)); C. Michael Haynes is credited with the cover figure.

2. **Magnetic Signatures of the Interaction Between Europa and Jupiter's Magnetosphere During the Juno Flyby**
   Peter Addison, **C. Michael Haynes**, Aaron Stahl, Lucas Liuzzo, Sven Simon, *Geophys. Res. Lett.*, 51(2), e2023GL106810, doi: 10.1029/2023GL106810, 2024. *All authors contributed equally to this study.*
   This article was selected as the cover story of the GRL issue (January 2024: Vol. 51(2)); C. Michael Haynes is credited with the cover figure.

1. **Emission of Energetic Neutral Atoms from the Magnetosphere-Atmosphere Interactions at Callisto and Europa**
   C. Michael Haynes, Tyler Tippens, Peter Addison, Lucas Liuzzo, Andrew R. Poppe, Sven Simon, *J.Geophys. Res. (Space Physics)*, 128(10), e2023JA031931, doi: 10.1029/2023JA031931, 2023

CONFERENCE PRESENTATIONS AND SEMINARS

(†) *Indicates an oral presentation.* ([I]) *Indicates an* ***invited*** *oral presentation.* ([P]) *Indicates a poster.*

17. † **C. M. Haynes**, S. Simon, and L. Liuzzo. **Detectability of ENA Emissions from the Magnetosphere-Atmosphere Interactions at Europa, Ganymede, and Callisto: Con-**

text for the JUICE Mission. *Magnetospheres of the Outer Planets Meeting*, Toulouse, France, 26-31 July, 2026.

16. [P] **C. M. Haynes**, S. Simon, and L. Liuzzo. **Impact of Consecutive Charge Exchange Interactions on the Emission of Energetic Neutral Atoms from Titan's Induced Magnetosphere**. *Magnetospheres of the Outer Planets Meeting*, Toulouse, France, 26-31 July, 2026.

15. [I] **C. M. Haynes**. **Emission of Energetic Neutral Atoms at Europa, Ganymede, and Callisto: Invited Seminar at JHU/APL**, (Virtual), 16 June, 2026.

14. [P] **C. M. Haynes**, S. Simon, and L. Liuzzo. **Emission of Energetic Neutral Atoms from Ganymede's Magnetosphere-Atmosphere Interaction**. *AGU Fall Meeting*, New Orleans, USA, 15-19 December, 2025.

13. [P] **C. M. Haynes**, S. Simon, and T. Tippens. **A Modeling Framework for the Analysis of ENA Images from Cassini's Titan Flybys**. *AGU Fall Meeting*, New Orleans, USA, 15-19 December, 2025.

12. [I] **C. M. Haynes**, S. Simon, and L. Liuzzo. **Detectability of ENA Emissions from the Magnetosphere-Atmosphere Interactions at Europa, Ganymede, and Callisto: Providing Context for the JUICE Mission (Invited)**. *AGU Fall Meeting*, New Orleans, USA, 15-19 December, 2025.

11. [P] **C. M. Haynes**, T. Tippens, S. Simon, and L. Liuzzo. **Observability of Energetic Neutral Atom Emissions at Europa and Callisto: Predictions for the JUICE Mission**. *AGU Fall Meeting*, Washington D.C., USA, 09-13 December, 2024.

10. [P] **C. M. Haynes**, T. Tippens, S. Simon, and L. Liuzzo. **Global Morphology and Detectability of ENA Emissions from the Magnetosphere-Atmosphere Interactions at Callisto and Europa**. *AGU Fall Meeting*, Washington D.C., USA, 09-13 December, 2024.

9. [†] **C. M. Haynes**, T. Tippens, S. Simon, and L. Liuzzo. **Observability of ENA Emissions at Europa and Callisto: Predictions for the JUICE Mission**. *EuroPlanet Science Congress*, Berlin, Germany, 08-13 September, 2024.

8. [P] **C. M. Haynes**, T. Tippens, P. Addison, L. Liuzzo, A. R. Poppe, and S. Simon. **Global Morphology of ENA Emissions from the Atmosphere-Magnetosphere Interactions at Callisto and Europa**. *EuroPlanet Science Congress*, Berlin, Germany, 08-13 September, 2024.

7. [P] **C. M. Haynes**, Peter Addison, Aaron Stahl, Lucas Liuzzo, and Sven Simon. **Magnetic Signatures of the Interaction Between Europa and Jupiter's Magnetosphere During the Juno Flyby.** *EuroPlanet Science Congress*, Berlin, Germany, 08-13 September, 2024.

6. [†] **C. M. Haynes**, T. Tippens, P. Addison, L. Liuzzo, A. R. Poppe, and S. Simon. **Global Morphology of ENA Emissions from the Atmosphere-Magnetosphere Interactions at Callisto and Europa**. *Magnetospheres of the Outer Planets Meeting*, Minneapolis, USA, 07-12 July, 2024.

5. [P] **C. M. Haynes**, T. Tippens, P. Addison, L. Liuzzo, A. R. Poppe, and S. Simon. **Observability of ENA Emissions at Europa and Callisto: Predictions for the JUICE Mission**. *Magnetospheres of the Outer Planets Meeting*, Minneapolis, USA, 07-12 July, 2024.

4. [P] **C. M. Haynes**, T. Tippens, P. Addison, L. Liuzzo, A. R. Poppe, and S. Simon. **Modeling the Detection of Energetic Neutral Atoms at Europa and Callisto: a Tool to Characterize Moon-Magnetosphere Interactions on a Global Scale**. *AGU Fall Meeting*, San Francisco, USA, 11-15 December, 2023.

3. [†] **C. M. Haynes**, T. Tippens, P. Addison, L. Liuzzo, A. R. Poppe, and S. Simon. **Emission of Energetic Neutral Atoms from the Magnetosphere-Atmosphere Interactions at**

Callisto and Europa. *AGU Fall Meeting*, San Francisco, USA, 11-15 December, 2023.

2. [P] **C. M. Haynes**, T. Tippens, P. Addison, L. Liuzzo, A. R. Poppe, and S. Simon. **Emission of Energetic Neutral Atoms at Callisto and Europa**. *AGU Fall Meeting*, Chicago, USA, 12-16 December, 2022.

1. [P] **C. M. Haynes**, T. Tippens, P. Addison, L. Liuzzo, A. R. Poppe, and S. Simon. **Emission of Energetic Neutral Atoms at Callisto and Europa**. *Magnetospheres of the Outer Planets Meeting*, Liége, Belgium, 11-15 July, 2022.

ADVISING

- **Troy Stephens (co-advised)** — Georgia Institute of Technology
  Undergraduate Student — August 2023 - January 2025
  Project: Using SPICE kernels to determine spacecraft position and bearing for Juno flybys of Jupiter's moons. Selected for Presidential Undergraduate Research Award (PURA) funding

- **Jeremiah Tunis (co-advised)** — Georgia Institute of Technology
  Undergraduate Student — May 2024 - August 2024
  REU Project: Tracing energetic test particles in Io's perturbed electromagnetic environment

- **Brendan McCluskey** — Georgia Institute of Technology
  Undergraduate Student — December 2022 - May 2023
  Project: Tracing the emission of hydrogen energetic neutral atoms from test particles at Europa

TEACHING

Advanced Space Plasma Physics — Atlanta, GA
Graduate Teaching Assistant — *January 2023 - May 2023*

- Teaching support, office hours, and homework/exam grading for a graduate course on the theoretical foundations of advanced plasma physics and their application throughout the solar system
- Topics include kinetic plasma theory, multi-fluid and magnetohydrodynamic treatments, cold plasma waves, shocks and discontinuities, planetary plasma interactions, and magnetospheric structure/topology

Co-Instructor/Co-Lecturer — *March 2025*

- Full-time lecturer for a week of class sessions; taught an overview of nonlinear MHD phenomena
- Provided derivation of Rankine-Huginot relations as well as a classification of shocks and discontinuities

Habitable Planets — Atlanta, GA
Graduate Teaching Assistant: Lecture — *May 2026 - August 2026*

- Grading, lecture moderation, and teaching contributions for an undergraduate course exploring the history of the universe, solar system, planetary/satellite formation and the evolution of Earth

Graduate Teaching Assistant: Laboratory — *May 2026 - August 2026*

- Administered weekly online lab meetings by delivering an introductory lecture exemplifying experimental or simulation results, and discussing material with students individually to ensure understanding

Introductory Physics I — Atlanta, GA
Undergraduate Teaching Assistant — *February 2020 - January 2022*

- Large undergraduate course on introductory mechanics, involving the supervision of weekly labs and the delivery of experimental and theoretical treatment in recitation-style lectures

HONORS AND AWARDS

Research Excellence Award April 2025
Awarded annually by the *GT School of Earth and Atmospheric Sciences* "for excellence in research, as judged primarily on the basis of the creativity and independence exhibited by the student in conduct of the research."

Best Paper Award April 2024
Awarded annually by the *GT School of Earth and Atmospheric Sciences* "for the best refereed paper or series of refereed papers, published by the time of selection, for which the student is the first author". Bestowed for *Emission of Energetic Neutral Atoms from the Magnetosphere-Atmosphere Interactions at Callisto and Europa* by *Haynes et al. (2023)*.

Outstanding Student Presenter Award (OSPA), AGU 2023 December 2023
Awarded at the American Geophysical Union (AGU) Fall Meeting for the talk: *Emission of Energetic Neutral Atoms from the Magnetosphere-Atmosphere Interactions at Callisto and Europa*. The award is given to the top 2%-5% of early career presenters, determined by senior scientists in the field.

Georgia Tech Presidential Fellowship August 2022 - Present
Awarded annually to the top 5% of the incoming class of graduate students at Georgia Tech. Fellowship includes an additional $5500 salary award and annual recognition at the Presidential Banquet.

UPC Silver Medal (contestant no. 434) November 2021
Awarded to the top 20% of contestants in the International University Physics Competition

Presidential Undergraduate Early Research Award (PURA): July 2018
Awarded to competitive research prospects within their first two years as undergraduates at Georgia Tech. Includes a $1500 salary award to fund a semester of original research.

Dean's List Fall 2018 - May 2022

Highest Honors Fall 2018 - December 2022

High Honors December 2022 - May 2022

PROFESSIONAL INVOLVEMENT

Oral Session Chair, Convener 2026
American Geophysical Union Fall Meeting (San Francisco)
Moon-Plasma Interactions Throughout the Solar System

Oral Session Chair, Convener 2024
American Geophysical Union Fall Meeting (Washington D.C.)
Moon-Plasma Interactions Throughout the Solar System

Poster Session Chair 2023
American Geophysical Union Fall Meeting (San Francisco)
Moon-Plasma Interactions Throughout the Solar System

American Geophysical Union November 2021 - Present
Member

PROFESSIONAL REFERENCES

Sven Simon, Professor Ph.D. Advisor
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Website: https://svensimon.gatech.edu/

Lucas Liuzzo, Research Scientist Collaborator
Space Sciences Laboratory, University of California, Berkeley
Email: liuzzo@berkeley.edu
Website: https://lukeliuzzo.github.io/

James Wray, Professor Qualfiying Exam & Thesis Defense Committee Member
School of Earth and Atmospheric Sciences
Georgia Institute of Technology
Email: jwray@gatech.edu
Phone: (404) 894-1992
Website: https://wray.eas.gatech.edu/

Mike Schatz, Professor Undergraduate Research Advisor
School of Physics, Georgia Institute of Technology
Email: mike.schatz@physics.gatech.edu
Website: https://schatzlab.gatech.edu/